

Here is the new standalone track, structured as its own unique path. It starts fresh at **Track 2: Module 1**, matching your existing formatting style perfectly while creating a clean break from the Raspberry Pi modules.

PC CORE INFRASTRUCTURE TRACK (STANDALONE ARCHITECTURE)

MODULE 1: THE BIOS HANDSHAKE & FIRMWARE NAVIGATION

Objective: Learn how to access low-level motherboard firmware, navigate system settings without a mouse, and safely exit without altering core system configurations.

Tech Briefing

Before any operating system (like Linux or Windows) can load, your computer runs a tiny, built-in program stored directly on a chip on the motherboard called the **BIOS** (Basic Input/Output System) or **UEFI**. The BIOS initializes the physical hardware components, checks that your RAM and processor are working, and decides which storage drive to hand control over to next.

Because the BIOS runs *before* the operating system, it has its own special command menu. However, you only have a split-second window to open it right when the physical power turns on.

Main Mission

1. **The Interception Window:** Shut down the Lenovo laptop completely. Turn it back on and watch the screen closely. Computer manufacturers assign a specific button to open the BIOS, but it only works if you tap it rapidly *immediately* after pressing the power button.
2. **The BIOS Probe:** If you miss the window and the computer boots up normally into your usual desktop, shut down the computer and try again. Try tapping one of these common diagnostic keys repeatedly right after pressing power: **F1**, **F2**, **F12**, or **Enter**. (*Hint: Lenovo ThinkPads usually use F1 or require you to press Enter to interrupt normal startup*).
3. **Firmware Navigation:** Once inside the BIOS utility, your mouse will not work. You must use only your keyboard **arrow keys** to shift between menus, and the **Enter** key to look inside options. Look across the top tabs and find the one labeled **Boot** or **Startup** to see where the computer lists the available storage drives.
4. **Safe Discard:** To ensure we don't accidentally alter how the computer functions, look at the bottom or side legends for the function key assigned to **Exit and Discard Changes** (usually **Esc** or **F10** followed by choosing "No"). Execute this command to restart the computer safely without saving any edits.

Side Quest

While exploring the main page of the BIOS, look for a section listing the hardware. Find the exact speed of your CPU (Processor) and the total amount of RAM (Memory) installed in this laptop. Write them down to show Dad.

MODULE 2: BOOTLOADERS & DUAL-BOOT NAVIGATION

Objective: Interrupt a bootloader countdown and navigate a GRUB menu to successfully jump between different operating systems.

Tech Briefing

When a computer has more than one operating system installed on its drive, it relies on a specialized piece of software called a **Bootloader**. On Linux systems, the standard bootloader is called **GRUB** (Grand Unified Bootloader).

GRUB acts like a traffic cop at a major intersection. When the computer turns on, it pauses the startup sequence, displays a menu of your available operating systems, and runs a countdown timer. If you don't touch anything, it automatically loads the

default choice (in this case, Ubuntu Linux). If you interrupt it, you can manually tell the hardware to route you into an entirely different environment, like Windows.

Main Mission

1. **The Countdown Interruption:** Turn on or restart the laptop. Wait for the dark **GRUB menu** to appear displaying your Ubuntu and Windows options. Look at the bottom of the screen to find the 10-second countdown timer. Before it hits zero, press the **Up** or **Down** arrow keys. Verify that touching the arrows freezes the countdown timer entirely.
2. **Windows Boot Execution:** Use your arrow keys to highlight **Windows Boot Manager** and press **Enter**. Let Windows load fully, log into the desktop, and explore for a moment.
3. **The Return Sequence:** Click the Windows Start Menu, select **Power**, and click **Restart**.
4. **Linux Boot Execution:** When the GRUB menu reapplies itself on restart, interrupt the timer again, but this time select **Ubuntu** and press **Enter**.
5. **The Multi-Boot Cycle:** Repeat this entire switching cycle (Booting to Windows, restarting, and booting back to Linux) two more times until you can comfortably toggle between both operating systems without hesitation.

Side Quest

While looking closely at the GRUB menu selections, look below the main Linux and Windows paths. Find the option labeled **Advanced options for Ubuntu** and navigate into it using your arrows. Look at the different version numbers listed. Ask Dad what a Linux "kernel" is and why GRUB keeps older versions safely stored away in this menu.